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Scallop migration is a parity break no down-gradient closure can produce: solver-measured phase-lag coefficients and a temperature-free field observable

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Ice and rock scallops migrate downstream as they evolve, a motion classically tied to the phase lag between the wall heat flux and the interface topography (Ashton & Kennedy 1972; Gilpin, Hirata & Cheng 1980; Hanratty 1981). A local, memoryless, down-gradient ('K-theory') closure carries no such lag. Freezing a sinusoidal ice–water interface in a penalized turbulence solver and projecting the time-averaged wall flux onto in-phase (amplitude) and quadrature (migration) harmonics, we measure every coefficient: the conduction amplitude rate is wavelength-independent (falsifying the K^2 and Mullins–Sekerka $|k|$ ansätze); the in-phase excess is negative at every amplitude and drive; and the quadrature grows sub-kinematically ($\propto U^{\approx 0.5}$, friction-velocity controlled), giving a damped, downstream-migrating mode. Swapping only the heat-transport operator at fixed momentum kills or misdirects the quadrature across a down-gradient closure battery: machine zero for a uniform eddy diffusivity, and at most 26% of resolved — with the wrong (upstream) sign — for flow-aware members. We construct a temperature-free observable $I = \tau\, c_{mig}/\lambda$ in which conductivity, latent heat and basal temperature drop cancel, computable from interface morphology alone; the solver pins $I = O(0.3–0.9)$ (motion-consistent value ≈ 0.16). A digitization of the Bushuk et al. (2019) adjustment-regime panels confirms the predicted downstream sign with net damping but places the magnitude at $I_{obs} \approx 10$ (estimator battery 5–17), the pre-registered anchoring-mismatch outcome for a train whose spacing formed under a 6× stronger pre-cut drive. A self-validated reanalysis harness (recovers I to $< 3\%$) awaits the raw $h(x, t)$ arrays.

1. Introduction

Scalloped erosion/melt patterns are ubiquitous on soluble and meltable surfaces — cave walls, glacier soffits, ice-shelf bases, meteorites, and lab ice (Curl 1966; Blumberg & Curl 1974; Ashton & Kennedy 1972; Bushuk et al. 2019). Their wavelength is fluid-selected through a friction-Reynolds criterion $Re_* = u_*\lambda/\nu \approx 2200$. A robust but under-exploited observation is that the pattern **migrates downstream** as it evolves — maximum ablation

Abstract must not spill onto p.2

sits downstream of the trough, at flow reattachment (Blumberg & Curl 1974; Gilpin, Hirata & Cheng 1980).

The migration direction and its phase-lag origin are established (Gilpin et al. 1980; Hanratty 1981): downstream migration corresponds to a flux–topography phase in $(\pi/2, \pi)$, and a local, memoryless, down-gradient closure carries no such phase. What is **new here** is not that fact but three things built on it: a phase-separated harmonic decomposition that returns the amplitude and migration rates as independent solver measurements; a direct operator-swap experiment that quantifies how completely a down-gradient closure discards the migration-carrying asymmetry; and a temperature-free morphological observable that makes the phase lag measurable from field geometry alone, without knowing the basal temperature drop or any material constant. The non-local, direction-selecting structure the migration requires is exactly the structure (memory, backscatter) that a local, down-gradient closure cannot represent.

Contributions.

1. A **phase-separated** wall-flux harmonic decomposition that returns the corrected mode $s = -\beta + i\omega_{mig}$ as an *output*, not an ansatz, with the conduction amplitude rate measured wavelength-independent — falsifying the $K^2/|k|$ curvature-smoothing ansätze for the conduction baseline, generalised across amplitude and drive (§3–§4).
2. A **direct operator-swap measurement** quantifying how far down-gradient closures fall short of the known phase-lag requirement: across a closure battery (uniform, Smagorinsky, wall-scaled mixing length) the migration signal collapses to machine zero or flips to the wrong (upstream) sign (§5).
3. A **temperature-free, constant-free morphological observable** $I = \tau \cdot c_{mig}/\lambda$, with an implemented, self-validated reanalysis harness and a regime-matched literature bound, that makes the phase lag measurable from field morphology alone (§6–§8).

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2. Method — harmonic decomposition of the wall flux

Freeze $y(x) = \bar{y} + a\sin(Kx)$ $(K = 2\pi n_w/L_x)$, time-average the per-column melt flux $m(x) = -\kappa\partial\theta/\partial y$, and project onto the two corrugation harmonics:

$E_{sin} = 2\langle e(x)\sin(Kx)\rangle$ (in-phase with the **shape** → amplitude change, β) $E_{cos} = 2\langle e(x)\cos(Kx)\rangle$ (in **quadrature** → pattern **migration**, ω_{mig})

E_{cos} is the Curl-1966 reattachment/lee signature: zero by symmetry unless the flow carries a direction. Two decompositions are run: the conduction baseline m_{cond} (flow off) for the $\beta \sim K^p$ scaling, and the flow-induced excess $e = m_{flow} - m_{cond}$ versus drive U. Freezing the interface removes boundary motion so the flux is uncontaminated; this is *why* the two coefficients separate cleanly, where a moving-boundary fit conflates them and returns unphysical (both-negative) coefficients.

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3. Findings

quantity	measured	ansatz it kills
conduction $\beta/a \sim K^{+0.13}$	wavelength-**independent**	K^2 curvature and Mullins–Sekerka $|k|$ **falsified**
in-phase flow excess E_{sin}	**negative at every driven** (a, U)	no autonomous $+\alpha a^{1/2}$ growth — **smoothing-limited**
quadrature E_{cos}	≈ 0 at $U = 0$, grows $\propto U^{0.5--0.8}$	migration is real and **friction-velocity** $(u_* \sim U^{1/2})$ controlled, not kinematic $(\propto U)$

Corrected mode: $s(K,U) = -\beta(K,U) + i \cdot \omega_{mig}(U)$, $Re(s) < 0$ always, $Im(s) \propto U^{1/2}$ — a damped, downstream-migrating pattern, not a growth–saturation balance.

Why this counts as a result, not a check. The corrected mode *fell out of* the decomposition; every coefficient is solver-measured; physical units are pinned independently by the Curl selection law without choosing λ by hand; and the final ratio is constant-free and falsifiable.

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4. Generalisation across amplitude and drive

4.1 Amplitude ($a_0/\lambda \in [0.05, 0.40]$). The two structural verdicts are not amplitude artefacts: the driven in-phase excess is **negative at every amplitude** (smoothing throughout, magnitude growing monotonically), and the conduction exponent stays $p \in [-0.27, +0.66]$, far below the K^2 (+2) and $|k|$ (+1) ansätze at every amplitude. *Honest caveat:* strict K-independence ($|p| < 0.6$) and amplitude-flat β/a hold only in the signal-rich regime $a_0/\lambda \gtrsim 0.2$; at shallow amplitude the single-wavenumber conduction signal is at the noise floor (the K^2/—k— falsification is robust regardless). (`amplitude_generalization_scan`.)

4.2 Drive (U to 6, beyond the swept $[1.5, 3.0]$). The genuine falsification risk is that strong-drive lee separation adds an *in-phase* component (an emergent growth term the weak-drive sweep would miss). It does not: E_{sin} is **negative at every** U out to $U = 6$, and the migration follows a **sub-kinematic** $E_{cos} \propto U^{+0.48}$ (≈½), far below the kinematic-advection alternative U^1. *Honest caveat:* migration is non-monotone — it peaks near $U \approx 4.5$ and rolls over by $U = 6$ (the lee structure reaches a limiting form), which *reinforces* the sub-kinematic reading. (`drive_window_scan`.)

U	in-phase E_{sin}	quadrature E_{cos}
0.0	-5.17×10^{-6} (smoothing)	$+4.9 \times 10^{-8}$ (≈0, parity control)
1.5	-6.23×10^{-5}	-6.59×10^{-5}
3.0	-9.19×10^{-5}	-1.14×10^{-4}
4.5	-8.86×10^{-5}	-1.38×10^{-4}
6.0	-4.81×10^{-5}	-1.19×10^{-4}

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5. A direct operator-swap measurement of the closure shortfall

A local, memoryless, down-gradient eddy diffusivity acting on a sinusoid is symmetric under $x \to -x$: it carries no flow direction and can only return an in-phase (real, damping) response — $E_{cos} \equiv 0$. This is the closure-space statement of the classical phase-lag requirement (Gilpin et al. 1980; Hanratty 1981). The resolved advective flux measured $E_{cos} \neq 0$, so migration requires the flow-direction-selecting, non-local reattachment asymmetry that K-theory discards. The contribution here is to make the shortfall **quantitative**: advancing the **same** momentum field at the **same** drive ($U = 3$, same interface) but transporting heat with the K-theory flux instead of resolved advection measures exactly how far each closure falls below the resolved migration:

heat transport (same $U = 3$, same interface)	E_{cos}	reading
resolved advective $-u \cdot \nabla\theta$	-1.14×10^{-4}	migrates
uniform eddy diffusivity $\nabla \cdot (K\nabla\theta)$	-1.2×10^{-18}	machine zero ($\sim 10^{-14}$ of resolved, $std = 0$)
flow-aware Smagorinsky $K_{eddy}(\|S\|)$	$+2.0 \times 10^{-6}$	$\sim 57\times$ below resolved, wrong (upstream) sign
wall-scaled mixing length $\ell = \kappa d$, $K = \ell^2\|S\|$	$+2.96 \times 10^{-5}$	$3.8\times$ below resolved, **wrong (upstream) sign**; outer-layer cap $\ell \le 0.2H$ changes it by $< 0.1\%$

All battery members still **smooth** ($E_{sin} \neq 0$): K-theory can damp but cannot migrate *downstream*. The direction-blind member is exactly parity-protected (machine zero); the flow-aware members inherit only the $|S|$-magnitude asymmetry of the resolved flow, which pushes their residual quadrature the *wrong way* (upstream, $\le$ 26% of resolved) — so no member of the down-gradient family reaches the observed downstream direction at any magnitude (`figures/59b_ktheory_closure_battery.json`). GPU/CuPy controls (Tesla P100) confirm (a) parity — $E_{cos} \to +2.1 \times 10^{-6}$ at $U = 0$ vs -8.4×10^{-5} at $U = 3$ (~40× smaller); and (b) the high-Re_* migration exponent tightens to 0.52 ($nx = 160$, four seeds, scatter ~1–2 %), on the friction-velocity $\frac{1}{2}$. (`scallop_ktheory_control.py`.)

This is the morphological analogue of the memory and backscatter signatures that a down-gradient closure omits in *flux* space: the closure's blindness does not merely lose information in the model — it leaves a fingerprint carved into the ice.

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6. A constant-free, $\mathit{\Delta}$T-free field test

A Stefan-number-free dimensional bridge ($\rho L v = q$, $q = -k_{th}(\mathit{\Delta}T/L_0)\partial\theta/\partial y$, length anchored by the corrugation wavelength) gives $\tau \sim \lambda^2/\mathit{\Delta}T$ (amplitude e-fold) and $c_{mig} \sim \mathit{\Delta}T/\lambda$ (migration speed). In the product the constants k_{th}, $\rho_i L$ **and** $\mathit{\Delta}T$ **cancel exactly**:

$I \equiv \tau \cdot c_{mig}/\lambda = Im(s)/(2\pi|Re(s)|)$ — the dimensionless ratio of migration rate to amplitude rate, basal-$\mathit{\Delta}T$-free.

The solver pins $I = O(0.3{-}{-}0.9)$ $(0.33, 0.68, 0.88$ at $n_w = 8, 12, 16$ — wavelength-dependent by a factor ≈ 2.7, an $O(1)$ morphological number rather than a universal constant).

Decoy warning (a methodological result). $\tau \propto \lambda^2/\mathit{\Delta}T$ *looks* like classic curvature/Mullins–Sekerka smoothing, but the λ^2 comes entirely from the conduction length↔time anchoring L_0^2/κ, **not** from a curvature operator — β/a is K-independent (§3). Fitting $\tau \propto \lambda^2$ in the field would *wrongly confirm* the very curvature ansatz the harmonic probe falsified; the only discriminator is the K-exponent of β/a (≈ 0 here vs $+2$ for curvature), which the dimensional scaling hides. Anyone analysing field scallops should measure the K-exponent, not fit $\tau \propto \lambda^2$.

Field-test protocol. On one scallop train measure λ (crest spacing), c_{mig} (downstream crest displacement between two visits), and τ (amplitude e-fold). Then:

outcome	reading
$I_{obs} \in [0.2, 1.0]$, c_{mig} downstream	consistent with $s = -\beta + i\omega_{mig}$
$c_{mig} \approx 0$ $(Im(s) = 0)$	parity-symmetric closure — K-theory **not** falsified on the ice (contradicts the solver)
c_{mig} upstream	new physics (e.g. depositional, not melt-limited)
$I_{obs} \gg 1$ or $\ll 0.1$	$\lambda \leftrightarrow u_*$ anchoring mismatch

Because I is $\mathit{\Delta}T$-free, a failure cannot be excused by an unknown basal $\mathit{\Delta}T$; measuring $c_{mig} + \lambda$ alone pins $\mathit{\Delta}T$, so τ becomes a prediction and the system is over-determined.

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7. Literature grounding

- λ **(wavelength) — solid.** $Re_* = u_*\lambda/\nu$ selection is mainstream: Blumberg & Curl (1974) ≈ 2200, Thomas (1979) ≈ 1000, Hsu–Locher–Kennedy (1979) $\lambda = 3180\nu/u_*$, Thorsness–Hanratty (1979)/Hanratty (1981) band 3100−−6300ν/u_*.
- c_{mig} **direction & mechanism — solid.** Downstream migration with maximum ablation downstream of the trough at reattachment: Curl (1966), Ashton & Kennedy (1972), Hsu et al. (1979), Gilpin et al. (1980). Gilpin et al.'s linear stability predicts downstream migration when the heat-flux/topography phase shift lies in $(\pi/2, \pi)$ — the same statement as $Im(s) \neq 0$.
- c_{mig} **magnitude and** τ **— sparse.** Cave-scallop surveys report λ only; migration speed and amplitude decay time are essentially never tabulated together for one train.

Two derived consistency relations with the turbulence closure (Paper 1). The migration quadrature $Im(s)$ that distinguishes a real closure is the imaginary part of a single complex transport admittance $Z = a_r + ia_i$, whose real part is the amplitude-rate / backscatter of the closure (Paper 1; Kraichnan 1976) and whose imaginary part is this scallop migration; a down-gradient eddy diffusivity is the projection $Z \mapsto \max(\mathrm{Re}\, Z, 0)$, which zeroes migration and backscatter *together*. Because the flux response is moreover causal, $\mathrm{Re}\, Z(k)$ and $\mathrm{Im}\, Z(k)$ are Kramers–Kronig (Hilbert) conjugates: any *scale-dependent* eddy viscosity — which every real closure (spectral eddy viscosity, Smagorinsky $\propto |S|$) is — **forces** a nonzero migration, so K-theory's pairing of a scale-dependent $\mathrm{Re}\, Z$ with $\mathrm{Im}\, Z \equiv 0$ is acausal (an $O(1)$ Kramers–Kronig residual). The measured migration $I = |\mathrm{Im}\, Z|/(2\pi|\mathrm{Re}\, Z|)$ is therefore not an incidental fact but the causality-mandated shadow of the same operator that carries backscatter in the turbulence closure — a derived link, verified on the admittance model (discrete Kramers–Kronig reconstruction of $\mathrm{Im}\, Z$ from $\mathrm{Re}\, Z$ to 5.3%).

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8. The one open gate — and a regime-matched bound

Sign caveat. Real ice ripples *grow* ($Re(s) > 0$, the Ashton–Kennedy/Gilpin/Hanratty moving-boundary instability), whereas the frozen-interface excess here is decay-only ($Re(s) < 0$) by construction. The migration $Im(s)$ is the robustly shared, K-theory-falsifying piece; matching the *growing* branch needs an interface-growth mechanism (a Röthlisberger opening) this smoothing-only solver does not contain.

The migration transfers to a moving boundary. We close the part of this gate the solver *can* reach. Seeding a single mode on the interface and advancing it under the Stefan melt feedback ($\rho L v = q$), the co-evolving interface tracks a **coherent damped-migrating eigenmode**: $log|Z|$ and $arg(Z)$ of the fundamental shape mode are both linear in time ($R^2 \geq 0.91$ for amplitude and phase, two seeds × two Stefan numbers). It **decays** ($Re(s) < 0$ at every St — the decay-only branch is *physical*, not an artefact of freezing) and **migrates downstream**, with $Im(s)$ tracking the frozen-probe quadrature E_{cos} (predicted $Im = E_{cos}/(aSt) \approx -0.43$ vs measured -0.40 at $St = 10^{-3}$, ˜5–10%). The observable $I = |Im|/(2\pi|Re|) \approx 0.15$−−0.17 is St**-invariant** (the rate scales as `1/St`, so $Re \cdot St$ and $Im \cdot St$ are constant to ˜3%), confirming I is a well-defined, normalization-free property of the mode. The migration — the closure-distinguishing quadrature — therefore **survives interface motion**. The total moving-interface decay runs $\sim 1.75\times$ faster than the flow-off conduction estimate (turbulence adds effective smoothing), so this motion-consistent $I \approx 0.16$ sits *below* the frozen flow-off band; the digitized Bushuk adjustment-train reading (below) lies *above* both — an anchoring mismatch of the observed train, not a solver discriminant. The *growing* branch ($Re(s) > 0$) is still out of reach here. (`scallop_moving_boundary_check.py`, `subglacial/moving_boundary_check.json`.)

A regime-matched real-data bound. Bushuk et al. (2019, *JFM* 873) track an evolving ice–water interface $h(x,t)$ at sub-mm/15 Hz through transition→equilibrium→**adjustment**. Their adjustment regime (after the drive was cut $U = 1.00 \rightarrow 0.16ms^{-1}$) **damps and migrates downstream simultaneously** — the exact analogue of the frozen probe ($s = -\beta + i\omega_{mig}$, $Re(s) < 0$, $Im(s) \neq 0$), so the sign caveat does not apply. We **digitized the adjustment-regime panels directly** from the openly hosted preprint (sha256-pinned; `glaciers/bushuk_fig3_digitize.py`: content-located page, tick-label-anchored piecewise calibration, density-tracked amplitude, slanted-comb crest fit; artifacts `figures/58b_bushuk_digitized.json/.png`). The digitized kinematics are: amplitude 11.0 → 9.2 mm over ∼ 300 min (early/late-window quartiles) — slow and **strongly oscillatory**, so τ comes from a fit-model battery: the physically-motivated damped-oscillator refit $A_0 e^{-t/\tau}[1 + b\cos(2\pi t/T + \phi)]$ gives $\tau = 2.6 \times 10^3$ min (68% CI 2.1–3.4×10^3; $T \approx 69$ min, $b \approx 0.03$, $R^2 = 0.42$), bracketed by envelope quartiles (1.2×10^3, conservative) and the log-linear fit (2.9×10^3, $R^2 = 0.25$; a two-exponential control degenerates, $R^2 = 0.30$); crest drift $c = 0.11$ (their published mean, anomalous periods excluded) to 0.17 mm min^{-1} (all-period comb); and the adjusting train's own spacing $\lambda = 27.9$ mm (6 crests, spacings 21–31.5 mm). An earlier by-eye reading ($\tau \approx 1$−−3 h, borrowed equilibrium-band $\lambda \approx 13$ cm, giving $I_{obs} \approx 0.10$) is **superseded** — its τ was $\geq 5\times$ too fast and its λ was not this train's spacing; the two errors happened to cancel toward the band:

> $I_{obs} = \tau \cdot c_{mig}/\lambda \approx 10$ (preferred damped-oscillator τ with the published c; estimator battery 4.9−−17): **downstream sign and net damping as predicted**, but a magnitude 6−−20× *above* the frozen flow-off band [0.33, 0.88] and 30–110× above the motion-consistent solver value 0.16 — i.e. squarely in the **pre-registered** $I_{obs} \gg 1$ *anchoring-mismatch* row of the §7 outcome table. That is the physically expected bin for this regime: the adjusting train retains the spacing formed under the *pre-cut* drive ($U = 1.00$ m s^{-1}) while damping under a 6× weaker flow, so its λ is anchored to the wrong u_* — exactly the mismatch the table names — and its oscillatory amplitude makes a single-eigenmode τ estimator-dependent (the dominant spread in I_{obs}). The **sign structure** — the closure-falsifying content — is confirmed; the **magnitude** question now rests on the raw-array pin: $h(x,t)$ through the *equilibrium* regime (where the solver band applies), read by the §6 harness, would replace both τ and c with the complex rate of the same mode. Candidate datasets beyond Bushuk's arrays (on request; co-authors D. Holland / A. Stern): Weady et al. (2022, PRL 128, 044502) interface extractions, and Cohen/Berhanu dissolution-scallop time-lapses (growth regime; sign caveat applies).

The pin. The one missing input is Bushuk's underlying $h(x,t)$ arrays, available from the authors on request (`mitchell.bushuk@noaa.gov`); the published JFM supplement is a parameter table only, not the interface time series, and no open data deposit exists. (The *equilibrium* regime cannot substitute for the raw arrays: there $Re(s) \approx 0$ by definition — the amplitude is statistically steady — so I is not readable from a published equilibrium panel; the adjustment regime is the only one whose figure exposes both rates, which is also why the raw $h(x,t)$ remains the true pin.) The §6 ratio is computed from morphology alone, and the reanalysis harness is **already implemented and self-validated**: `harmonic_mode_rate(x,t,H)` rFFTs each frame, tracks the dominant mode $a_k(t)e^{i\phi_k(t)}$, fits $Re(s) = d/dt \ln a_k$, $Im(s) = d\phi_k/dt$, and returns I plus the downstream sign; on a synthetic damped, downstream-migrating train with **known** $(Re(s), c_{mig}, \lambda)$ it recovers all three to $< 2\%$ and I to $< 3\%$. It ingests Bushuk's arrays directly — no further code, just the data. (`scallop_field_test.py`.)

9. Discussion and scope

This is a solver-verified, quantitative characterization of the closure: the corrected mode, the operator-swap shortfall, and the temperature-free ratio are all self-consistent and GPU-reproducible. **Dimensionality.** All probes here are 2-D (x–y), while real scallops are three-dimensional cups with spanwise structure. The *structural* verdicts are dimension-robust by symmetry: a local down-gradient closure carries no flow direction in any dimension, so its quadrature vanishes (or, flow-aware, misdirects) in 3-D exactly as in §5 — the parity argument never references dimensionality. The *numbers* (the $U^{\approx 0.5}$ quadrature growth, the $K^{+0.13}$ conduction exponent, the solver band for I) are 2-D measurements; their 3-D counterparts are part of the field-pin step, not assumed. It is **not yet validated against field scallops** — §8 is precisely why that test is cheap (no ΔT needed). The banked claims are the qualitative three — $Im(s) \to 0$ at $U = 0$, strictly sub-linear migration, exponent $\to \frac{1}{2}$ at high Re_* (central value 0.52) — robust to the ˜±0.1 scatter from spin-up/seed/U-grid. The solver-side moving-boundary measurement (§8) is now done — the migration transfers to an actually-moving interface and I is motion-invariant — so what a field dataset would still pin is the migration exponent's precise value and the field I_{obs} from real $h(x, t)$.

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10. Reproduce

```
python glaciers/scallop_amplitude_harmonics.py # §3–§4; writes figures/56_*.json
python glaciers/scallop_ktheory_control.py # §5 keystone control (CPU; xp=cupy for
↪  GPU)
python glaciers/scallop_field_test.py # §6/§8; figures/58_*.json (harness
↪  self-check + superseded figure bound)
python glaciers/bushuk_fig3_digitize.py # §8; figures/58b_*.json (digitized exp-1b
↪  kinematics)
python glaciers/scallop_moving_boundary_check.py # §8 moving-boundary migration
↪  transfer + I(St-invariance)
pytest glaciers/tests/test_scallop_amplitude_harmonics.py \
       glaciers/tests/test_scallop_ktheory_control.py \
       glaciers/tests/test_scallop_field_test.py
```

Data and code availability

All scallop-migration and harmonic-decomposition results regenerate from the scripts in the reproduce section; the field-test harness and its synthetic validation data are in the public repository at https://github.com/abdh0saifelden2-debug/K.

References

Ashton, G. D., & Kennedy, J. F. (1972). Ripples on underside of river ice covers. *Journal of the Hydraulics Division (ASCE)* 98, 1603–1624. doi:10.1061/JYCEAJ.0003407

Blumberg, P. N., & Curl, R. L. (1974). Experimental and theoretical studies of dissolution roughness. *Journal of Fluid Mechanics* 65, 735–751. doi:10.1017/S0022112074001625

Bushuk, M., Holland, D. M., Stanton, T. P., Stern, A., & Gray, C. (2019). Ice scallops: a laboratory investigation of the ice–water interface. *Journal of Fluid Mechanics* 873, 942–976. doi:10.1017/jfm.2019.398

Curl, R. L. (1966). Scallops and flutes. *Transactions of the Cave Research Group of Great Britain* 7, 121–160.

Gilpin, R. R., Hirata, T., & Cheng, K. C. (1980). Wave formation and heat transfer at an ice-water interface in the presence of a turbulent flow. *Journal of Fluid Mechanics* 99, 619–640. doi:10.1017/S0022112080000791

Hanratty, T. J. (1981). Stability of surfaces that are dissolving or being formed by convective diffusion. *Annual Review of Fluid Mechanics* 13, 231–252. doi:10.1146/annurev.fl.13.010181.0013

Hsu, K.-H., Locher, F. A., & Kennedy, J. F. (1979). Forced-convection heat transfer from irregular melting wavy boundaries. *Journal of Heat Transfer* 101, 598–602. doi:10.1115/1.3451043

Mullins, W. W., & Sekerka, R. F. (1964). Stability of a planar interface during solidification of a dilute binary alloy. *Journal of Applied Physics* 35, 444–451. doi:10.1063/1.1713333

Thomas, R. M. (1979). Size of scallops and ripples formed by flowing water. *Nature* 277, 281–283. doi:10.1038/277281a0

Thorsness, C. B., & Hanratty, T. J. (1979). Mass transfer between a flowing fluid and a solid wavy surface. *AIChE Journal* 25, 686–697. doi:10.1002/aic.690250415